A Transfer Learning and Hybrid Classification Approach for AI-Generated Images Detection

Amitoj Nayar
Dublin City University
amitoj.nayar2@mail.dcu.ie

Thuso Sehloho
Dublin City University

***Abstract*—The growing usage of AI-generated images on e-commerce platforms creates significant risks to the integrity of the marketplace and consumer trust. This report evaluates several transfer learning and machine learning techniques for the binary classification of images as authentic or AI-generated. Following a comparative analysis of various backbones and classification heads, this study identifies a Hybrid CNN-SVM approach as the optimal solution. By leveraging EfficientNetV2-S fine-tuned on a small labelled dataset and employing a Radial Basis Function (RBF) Kernel Support Vector Machine (SVM), the system achieves a final Test F1 score of 0.908 ± 0.014, providing a scalable solution for AI-generated content detection in commercial ecosystems.**

I. Introduction

The digital commerce landscape is currently going through a crisis of authenticity caused by the democratization of high-fidelity generative models. As Generative Adversarial Networks (GANs) and Diffusion Models (DMs) achieve unprecedented levels of photorealism, the capacity for sellers to generate misrepresenting product listings has increased exponentially.

Algorithmic intervention is no longer just a research interest but a commercial necessity. Modern GAN-generated images have reached a level where they are often indistinguishable from real photos to the human eye, and in some cases, judged as more trustworthy [1]. Recent studies indicate that human accuracy in detecting diffusion-based synthetic images is close to random guessing.

While traditional digital forensics historically relied on physical sensor inconsistencies (such as PRNU), modern generative techniques produce imagery that lacks these mechanical signatures. Consequently, forensic tools must evolve to identify subtle algorithmic fingerprints embedded within high-dimensional feature representations.

Through an evaluation of various techniques, this report details the implementation of a hybrid EfficientNetV2-SVM architecture aiming to restore a layer of trust in digital media.

II. Related Work

A. Transfer Learning and Inductive Biases

Transfer learning has become the standard approach in image classification tasks where labeled data is limited. By leveraging pretrained feature representations, models can achieve strong performance with minimal fine-tuning. Traditionally, ImageNet-1k was the standard for pretraining. However, recent research has highlighted the significant benefits of pretraining on the larger and more diverse ImageNet-21k dataset [2]. This dataset, containing approximately 14 million images and 21,000 classes, enables even small models like EfficientNetV2-S to develop more robust and generalizable features.

Furthermore, the choice of architecture family is critical in transfer learning. Recent studies demonstrate that Convolutional Neural Networks (CNNs) can generalize better with smaller datasets and get better accuracy than Vision Transformers (ViTs) [3]. Consequently, for a limited dataset, CNNs provide a more robust foundation than transformer-based alternatives.

B. Hybrid CNN-SVM Architectures

The integration of deep learning with SVMs has been explored to improve generalization in restricted data regimes. Traditional Softmax layers utilize Empirical Risk Minimization (ERM), which focuses on maximizing the log-likelihood of the training data. In contrast, SVMs utilize the Structural Risk Minimization (SRM) principle to maximize the margin between classes, enhancing robustness against outliers. Research has shown that replacing the Softmax head with an SVM can provide small but consistent performance gains on popular datasets like MNIST and CIFAR-10. Recent applications in medical imaging have validated the hybrid CNN-SVM approach as an optimal solution for high-dimensional feature vectors [4].

III. Methodology

The methodology for this project was designed to maximize the forensic signal-to-noise ratio while ensuring high generalization under the constraint of a small dataset. The pipeline encompasses data preprocessing, architectural selection, optimization strategy, feature extraction, and the implementation of the hybrid classification head.

Figure 1 illustrates the proposed hybrid architectural pipeline, detailing the progression from image input and fine-tuned EfficientNetV2-S feature extraction to the final 1280-dimensional embedding classification via an RBF-Kernel SVM.

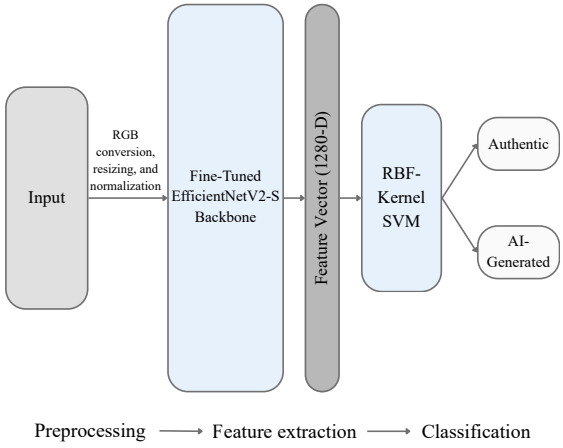


Fig. 1. Architectural overview of the proposed hybrid framework

A. Data Preprocessing and Augmentation

Due to computational resource constraints, the model was evaluated across five experimental runs utilizing a stratified ten-fold cross-validation framework. This approach ensured that 50% of the total dataset served as unseen test data across the iterations. While a full ten-fold rotation was not feasible, this five-fold subset remains statistically rigorous, providing a sufficient sample size to demonstrate the stability of the hybrid EfficientNetV2-SVM pipeline. For each experimental run, the dataset was partitioned using a stratified 80/10/10 split, resulting in training, validation, and test subsets. This split ensures that the class balance (approximately 52% authentic to 48% AI-generated) is maintained across all subsets.

To simulate real-world marketplace degradation, a data augmentation pipeline was implemented. Marketplace images are frequently subjected to social media processing, which involves JPEG compression and resizing. The augmentation strategy included Gaussian blurring, JPEG compression (quality factors 60-95), geometric transformations through horizontal and vertical flips, and color jittering.

B. Fine-Tuning Strategy

The EfficientnetV2-S.in21k model [5] was selected as the core backbone for this task.

Fine-tuning was conducted over 25 epochs (with early stopping) using the AdamW optimizer [6], which improves generalization by decoupling weight decay from the adaptive learning rate updates. The initial learning rate was set to $\eta = 5 \times 10^{-5}$ with a weight decay coefficient of $\lambda = 1 \times 10^{-2}$. To stabilize convergence, a Cosine Annealing learning rate schedule [7] was applied. The learning rate followed a half-cosine curve decaying to a minimum of $\eta_{\min} = 1 \times 10^{-6}$.

C. Feature Extraction

Following fine-tuning, the EfficientNetV2-S backbone functions as a high-dimensional feature extractor. For every image in the dataset, the extraction process involves (1) conversion to the standard RGB format, (2) resizing to a consistent input resolution and normalization using the pre-training statistics (mean and standard deviation) of ImageNet-21k, (3) output extraction from the penultimate layer (Global Average Pooling), resulting in a feature vector with a dimension of 1280.

D. Hybrid Classification

The final classification is performed by a RBF-Kernel SVM ($C = 2$) trained on the extracted features. SVMs are highly effective for high-dimensional classification on small datasets as they maximize the margin between classes, enhancing robustness and generalization.

By using the CNN as a feature generator and the SVM as the discriminator, the hybrid model exploits the strengths of both deep learning and classical statistical learning theory.

IV. Experiments & Results

A. Dataset

The data used consists of labelled training and unlabelled test sets of product-listing images. Training data includes images with binary ground truth labels (authentic or AI-generated) whereas for the test set, the ground truth is not provided. Therefore, all evaluation metrics in this paper are on the training set.

B. Implementation details

The experiments were made on Google Colab with a T4 GPU, using the timm library for the pre-trained models and torchvision for data augmentation and preprocessing. Feature classification and hyperparameter optimization were conducted using scikit-learn.

C. Experiments

1) EfficientNetV2-S Feature Extraction: Initial benchmarking focused on using a frozen EfficientNetV2-S backbone pre-trained on ImageNet-21k as a static feature extractor. A comparison was conducted between two classification heads: SVM and Logistic Regression. The SVM achieved a superior F1 score of 0.877, slightly outperforming the Logistic Regression with a F1 score of 0.871.

2) Multi-Model Feature Concatenation: To explore the potential of ensemble representations, features from DinoV2 and ConvNeXt-V2 were extracted and concatenated into a singular 2048-dimensional vector. Despite the increased complexity and diversity of the feature space, the SVM classification resulted in a lower F1 score of 0.841. This performance dip suggested that simple feature fusion without domain-specific fine-tuning introduced excessive noise for this specific dataset.

3) ResNet50 and LightGBM: One of the models explored was a fine-tuned ResNet50 with a LightGBM classifier. Training for this initial model involved five epochs, during which the training loss dropped rapidly from 0.6127 to 0.0356. However, validation loss began to increase after Epoch 2, moving from 0.4024 to 0.5806 by the final epoch, indicating

early overfitting. The model achieved the F1 score of 0.804, outperforming the baseline LightGBM at 0.799 and the individual fine-tuned ResNet50 model at 0.793.

To improve the F1 score of 0.804, several advancements have been implemented. The training data augmentation strategy was enhanced to improve model robustness and generalization. This enhancement involved introducing geometric transformations through crops and rotations and making color jittering variations more aggressive. Following the implementation of the enhanced augmentation, the model's F1 score on the validation set was 0.722. The immediate next step was to implement hyperparameter tuning for the LightGBM classifier, which led to a final 0.830 F1 score.

4) Fine-Tuned Hybrid Classifier: Based on the superior performance of EfficientNetV2-S compared to the DinoV2, ConvNeXt-V2, and ResNet50 configurations, this architecture was selected as the primary backbone for end-to-end fine-tuning. Over 15 epochs, the model showed consistent convergence, with the training loss reaching 0.468 and the validation loss settling at 0.338. This approach improved the validation F1 score to 0.881, confirming that adapting the backbone's internal representations to the marketplace dataset was more effective than using generic pre-trained features. The Uniform Manifold Approximation and Projection (UMAP) plot (Figure 2) comparing the 1280-dimensional embeddings from the frozen ImageNet-21k backbone (a) versus the fine-tuned backbone (b) reveals a distinct transformation.

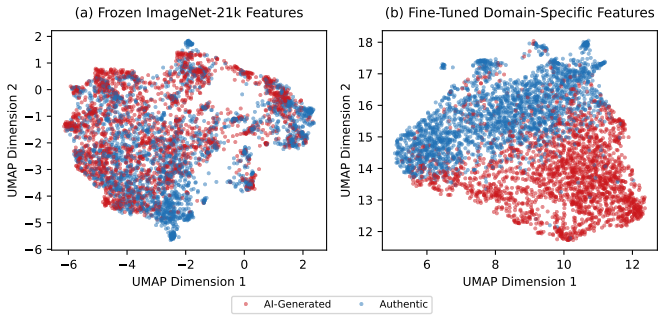


Fig. 2. Comparative UMAP visualization of 1280-dimensional feature embeddings

Building on the success of the fine-tuned backbone, the standard Softmax output head was replaced with an SVM. By using the learned representations from the fine-tuned EfficientNetV2 as inputs for the SVM, the F1 score went up to 0.890.

5) Augmented Hybrid Optimization: The final experimental phase integrated the data augmentation pipeline described in the methodology section. This led to a significant reduction in training loss (0.306) and validation loss (0.248), achieving a validation F1 score of 0.897.

A final comparison was conducted between SVM and XGBoost using these optimized features, where the SVM reached the project's peak performance with a F1 score of 0.908 ± 0.014. The aggregate performance metrics for these top-performing configurations are summarized in Table I.

TABLE I
AGGREGATE PERFORMANCE COMPARISON (MEAN $\pm$ STD) OVER FIVE STRATIFIED FOLDS

Model	F1 Score	Recall	AUC-ROC
XGBoost	0.900 ± 0.017	0.902 ± 0.027	0.964 ± 0.010
SVM	$\mathbf{0.908 \pm 0.014}$	$\mathbf{0.918 \pm 0.016}$	$\mathbf{0.967 \pm 0.009}$

D. Results

Table II provides an overview of the F1 scores achieved across the various configurations explored during the project.

TABLE II
SUMMARY OF CLASSIFICATION PERFORMANCE ACROSS EXPERIMENTAL CONFIGURATIONS

Method / Architecture	F1 Score
Baseline (Random Forest)	0.636
ResNet50 (Fine-tuned) + LightGBM	0.804
ResNet50 (Fine-tuned + Augm.) + LightGBM	0.830
DinoV2 + ConvNeXt-V2 (Frozen Concatenation) + SVM	0.841
EfficientNetV2-S (Frozen) + Logistic Regression	0.871
EfficientNetV2-S (Frozen) + SVM	0.877
EfficientNetV2-S (Fine-tuned)	0.881
EfficientNetV2-S (Fine-tuned) + SVM	0.890
EfficientNetV2-S (Fine-tuned + Augm.)	0.897
EfficientNetV2-S (Fine-tuned + Augm.) + XGBoost	0.900
EfficientNetV2-S (Fine-tuned + Augm.) + SVM	**0.908**

To provide a detailed visualization of the model's diagnostic behavior, the following figures display the results from Fold 1, which was specifically selected as it represents the median performance across the 5-fold cross-validation process.

The confusion matrix (Figure 3) reveals a balanced error distribution. In an online marketplace context, minimizing False Positives (Authentic images misclassified as AI) is critical to avoid suppressing legitimate merchants, while low False Negatives (AI images misclassified as Authentic) are essential for maintaining platform integrity. Specifically, the model demonstrates strong precision and recall with only 22 False Positives and 19 False Negatives. This balance between false positives and negatives suggests the hybrid architecture is well-tuned, maintaining a high detection rate while minimizing unnecessary merchant suppression.

The diagnostic robustness of the final SVM model was evaluated via a Receiver Operating Characteristic (ROC) curve. The model achieved an Area Under the Curve (AUC) of 0.967, as shown in Figure 4. This high score reinforces the model's reliability, proving it is robust enough to handle the subtle nuances between real-world marketplace photos and high-fidelity AI generations.

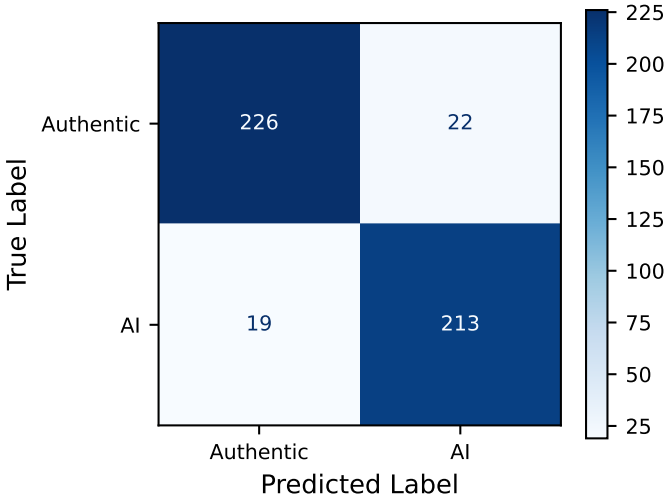


Fig. 3. Confusion Matrix of the final model.

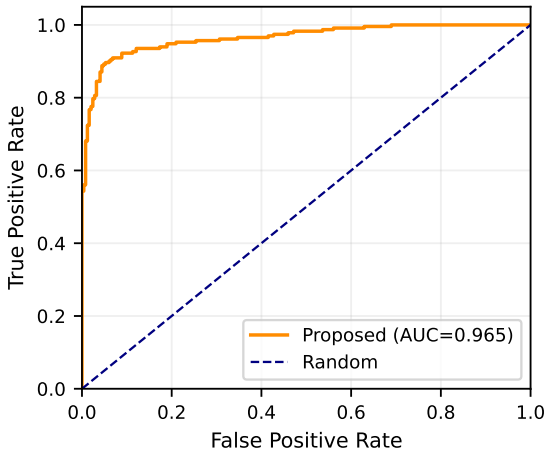


Fig. 4. ROC Curve of the final model.

V. Discussions

The achievement of a 0.908 $\pm$ 0.014 F1 score represents a 42.7% relative improvement over the baseline baseline model of 0.636, validating the rigor of the proposed hybrid architecture.

However, a qualitative inspection of the model's failure cases reveals that high discriminative power does not necessarily mean that the model is perfect. As seen in qualitative error analysis, the model still produces high-confidence misclassifications. Authentic images containing high-frequency textures or sensor noise can occasionally mimic the amplitude spectrum fingerprints typical of DMs, leading to False Positives.

Future work should investigate ensemble methods that incorporate frequency-domain features directly into the network architecture to improve detection of high-frequency Diffusion Model artifacts. Additionally, exploring Vision Transformers as backbones for larger datasets could provide further insights into the scalability of this hybrid approach.

VI. Conclusion

The detection of AI-generated images in e-commerce requires an approach that accounts for both the sophistication of generative models and the practical limitations of real-world data. This report has demonstrated that a hybrid architecture, combining the feature extraction of a fine-tuned EfficientNetV2 backbone with the decision boundary of an RBF-Kernel SVM, can effectively distinguish between authentic and AI-generated content with high precision. The integration of AdamW optimization and Cosine Annealing ensured stable convergence, while the data augmentation strategy provided a necessary defense against common image degradations.

The findings emphasize the power of transfer learning for small-to-medium datasets. The specific architectural artifacts of GANs and DMs are best captured through task-specific fine-tuning. Furthermore, the hybrid SVM head proves to be a better alternative to Softmax for binary classification tasks where generalization is important.

Ultimately, this framework provides a rigorous foundation for rebuilding digital trust in an era where the boundary between reality and synthesis continues to blur.

Contribution

Amitoj worked on the EfficientNetV2-SVM pipeline. He also tested this against other models like XGBoost and Logistic Regression to prove which worked best. Additionally, he ran experiments combining features from DinoV2 and ConvNeXt-V2.

Thuso worked on the ResNet50-LightGBM pipeline, incorporating data augmentation, threshold optimization, and hyperparameter tuning. He also explored an upgrade using EfficientNetB4.

Both of us collaborated on writing and finalizing this report.

References

[1] Sophie Nightingale and Hany Farid. *AI-synthesized faces are indistinguishable from real faces and more trustworthy*, Proceedings of the National Academy of Sciences, Vol. 119, 2022.

[2] Tal Ridnik, Emanuel Ben-Baruch, Asaf Noy and Lihi Zelnik-Manor. *ImageNet-21K Pretraining for the Masses*, arXiv:2104.10972, 2021.

[3] José Maurício, Ines Domingues, Jorge Bernardino. *Comparing Vision Transformers and Convolutional Neural Networks for Image Classification: A Literature Review*, Applied Sciences, Vol. 13, 2023

[4] Linjie Qiu and Jian Zhai. *A hybrid CNN-SVM model for enhanced autism diagnosis*, PLoS ONE, vol. 19, no. 5, 2024

[5] Mingxing Tan and Quoc V. Le. *EfficientNetV2: Smaller Models and Faster Training*, arXiv:2104.00298, 2021.

[6] Ilya Loshchilov and Frank Hutter. *Decoupled Weight Decay Regularization*, arXiv:1711.05101, 2019.

[7] Ilya Loshchilov and Frank Hutter. *SGDR: Stochastic Gradient Descent with Warm Restarts*, arXiv:1608.03983, 2017.